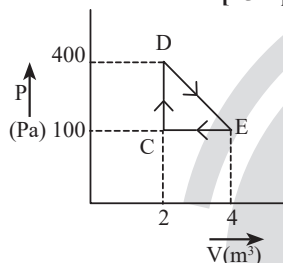


Thermodynamics and Kinetic Theory of Gases

Single Correct Type Questions

1. A thermodynamic system is taken through cyclic process. The total work done in the process is:

[15 Apr, 2023 (Shift-I)]



- (a) 100 J (b) 300 J (c) Zero (d) 200 J
2. 1 kg of water at 100°C is converted into steam at 100°C by boiling at atmospheric pressure. The volume of water changes from $1.00 \times 10^{-3} \text{ m}^3$ as a liquid to 1.671 m^3 as steam. The change in internal energy of the system during the process will be (Given latent heat of vaporisation = 2257 kJ/kg . Atmospheric pressure = $1 \times 10^5 \text{ Pa}$)

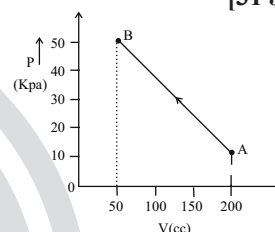
[11 Apr, 2023 (Shift-I)]

- (a) + 2090 kJ (b) - 2090 kJ
(c) - 2426 kJ (d) + 2476 kJ
3. Given below are two statements:
Statement-I: If heat is added to a system, its temperature must increase.
Statement-II: If positive work is done by a system in a thermodynamic process, its volume must increase.
In the light of the above statements, choose the correct answer from the options given below

[8 Apr, 2023 (Shift-I)]

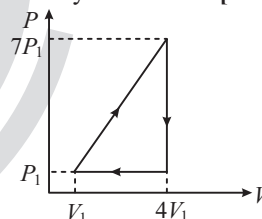
- (a) Statement I is true but Statement II is false
(b) Both Statement I and Statement II are true
(c) Both Statement I and Statement II are false
(d) Statement I is false but Statement II is true

4. The pressure of a gas changes linearly with volume from A to B as shown in figure. If no heat is supplied to or extracted from the gas then change in the internal energy of the gas will be [31 Jan, 2023 (Shift-I)]



- (a) 6 J (b) Zero (c) -4.5 J (d) 4.5 J

5. In the cyclic process shown in the figure, the work done by the gas in one cycle is [30 Jan, 2023 (Shift-I)]



- (a) $28 P_1 V_1$ (b) $14 P_1 V_1$ (c) $18 P_1 V_1$ (d) $9 P_1 V_1$

6. Heat is given to an ideal gas in an isothermal process.
A. Internal energy of the gas will decrease.
B. Internal energy of the gas will increase.
C. Internal energy of the gas will not change.
D. The gas will do positive work.
E. The gas will do negative work.

Choose the correct answer from the options given below:
[30 Jan, 2023 (Shift-I)]

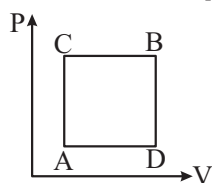
- (a) A and E only (b) B and D only
(c) C and E only (d) C and D only

7. A cylinder of fixed capacity of 44.8 litres contains helium gas at standard temperature and pressure. The amount of heat needed to raise the temperature of gas in the cylinder by 20.0°C will be: [29 June, 2022 (Shift-I)]
(Given gas constant $R = 8.3 \text{ JK}^{-1} \text{ mol}^{-1}$)

- (a) 249 J (b) 415 J (c) 498 J (d) 830 J

8. A gas can be taken from A to B via two different processes ACB and ADB. When path ACB is used 60 J of heat flows into the system and 30 J of work is done by the system. If path ADB is used work done by the system is 10 J. the heat flow into the system in path ADB is

[9 Jan, 2019 (Shift-I)]



- (a) 40 J (b) 80 J (c) 100 J (d) 20 J

9. A sample of gas at temperature T is adiabatically expanded to double its volume. The work done by the gas in the process is (given, $\gamma = \frac{3}{2}$): [01 Feb, 2023 (Shift-I)]

- (a) $W = TR[\sqrt{2} - 2]$ (b) $W = \frac{T}{R}[\sqrt{2} - 2]$
(c) $W = \frac{R}{T}[2 - \sqrt{2}]$ (d) $W = RT[2 - \sqrt{2}]$

10. Consider two containers A and B containing monoatomic gases at the same Pressure (P), Volume (V) and Temperature (T). The gas in A is compressed isothermally to $\frac{1}{8}$ of its original volume while the gas B is compressed adiabatically to $\frac{1}{8}$ of its original volume. The ratio of final pressure of gas in B to that of gas in A is:

[10 Apr, 2023 (Shift-I)]

- (a) 8 (b) $\frac{3}{8^2}$ (c) $\frac{1}{8}$ (d) 4

11. Match List-I with List-II:

List-I		List-II	
A.	Isothermal Process	I.	Work done by the gas decreases internal energy
B.	Adiabatic Process	II.	No change in internal energy
C.	Isochoric Process	III.	The heat absorbed goes partly to increase internal energy and partly to do work
D.	Isobaric Process	IV.	No work is done on or by the gas

Choose the correct answer from the options given below:

[25 Jan, 2023 (Shift-II)]

- (a) $A \rightarrow \text{II}; B \rightarrow \text{I}; C \rightarrow \text{III}; D \rightarrow \text{IV}$
(b) $A \rightarrow \text{II}; B \rightarrow \text{I}; C \rightarrow \text{IV}; D \rightarrow \text{III}$
(c) $A \rightarrow \text{I}; B \rightarrow \text{II}; C \rightarrow \text{IV}; D \rightarrow \text{III}$
(d) $A \rightarrow \text{I}; B \rightarrow \text{II}; C \rightarrow \text{III}; D \rightarrow \text{IV}$

12. Under isothermal condition, the pressure of a gas is given by $P = aV^{-3}$, where a is a constant and V is the volume of the gas. The bulk modulus at constant temperature is equal to

[13 Apr, 2023 (Shift-I)]

- (a) $P/2$ (b) $3P$ (c) $2P$ (d) P

13. A bubble has surface tension S . The ideal gas inside the bubble has ratio of specific heats $\gamma = \frac{5}{3}$. The bubble is exposed to the atmosphere and it always retains its spherical shape. When the atmospheric pressure is P_{a1} , the radius of the bubble is found to be r_1 and the temperature of the enclosed gas is T_1 . When the atmospheric pressure is P_{a2} , the radius of the bubble and the temperature of the enclosed gas are r_2 and T_2 , respectively.

[JEE Advance 2022]

Which of the following statement(s) is(are) correct?

- (a) If the surface of the bubble is a perfect heat insulator,

$$\text{then } \left(\frac{r_1}{r_2}\right)^5 = \frac{P_{a2} + \frac{2S}{r_2}}{P_{a1} + \frac{2S}{r_1}}$$

- (b) If the surface of the bubble is a perfect heat insulator, then the total internal energy of the bubble including its surface energy does not change with the external atmospheric pressure.

- (c) If the surface of the bubble is a perfect heat conductor and the change in atmospheric temperature is

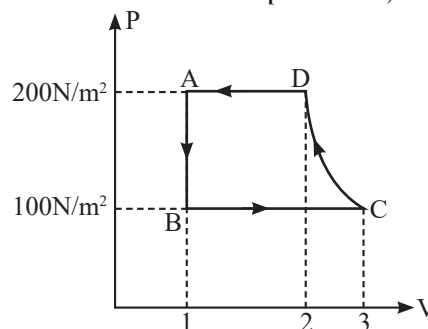
$$\text{negligible, then } \left(\frac{r_1}{r_2}\right)^3 = \frac{P_{a2} + \frac{4S}{r_2}}{P_{a1} + \frac{4S}{r_1}}$$

- (d) If the surface of the bubble is a perfect heat insulator,

$$\text{then } \left(\frac{T_2}{T_1}\right)^{\frac{5}{2}} = \frac{P_{a2} + \frac{4S}{r_2}}{P_{a1} + \frac{4S}{r_1}}$$

14. The P-V diagram of a diatomic ideal gas system going under cyclic process as shown in figure. The work done during an adiabatic process CD is (use $\gamma = 14$):

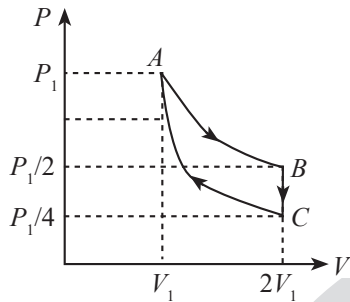
[18 March, 2021 (Shift-I)]



- (a) -500J (b) 400J
(c) -400J (d) 200J

15. If one mole of an ideal gas at (P_1, V_1) is allowed to expand reversibly and isothermally ($A \rightarrow B$) its pressure is reduced to one-half of the original pressure (see figure). This is followed by a constant volume cooling till its pressure is reduced to one-fourth of the initial value ($B \rightarrow C$). Then it is restored to its initial state by a reversible adiabatic compression ($C \rightarrow A$). The net work done by the gas is equal to:

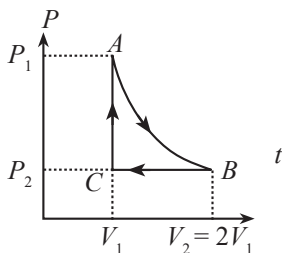
[24 Feb, 2021 (Shift-II)]



- (a) 0 (b) $RT \ln 2$
(c) $RT \left(\ln 2 - \frac{1}{2(\gamma-1)} \right)$ (d) $-\frac{RT}{2(\gamma-1)}$
16. n mole of perfect gas undergoes a cyclic process $ABCA$ (see figure) consisting of the following processes.
 $A \rightarrow B$: Isothermal expansion at temperature T so that the volume is doubled from V_1 to $V_2 = 2V_1$ and Pressure changes from P_1 to P_2 .
 $B \rightarrow C$: Isobaric compression at pressure P_2 to initial volume V_1 .
 $C \rightarrow A$: Isobaric change leading to change of pressure from P_2 to P_1 .

Total work done in the complete cycle $ABCA$ is:

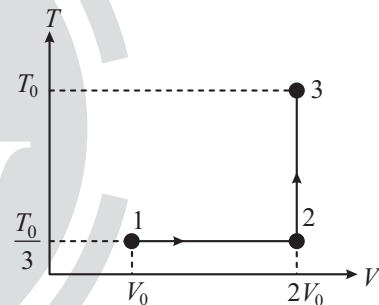
[24 Feb, 2021 (Shift-I)]



- (a) $nRT \left(\ln 2 + \frac{1}{2} \right)$
 (b) $nRT \left(\ln 2 - \frac{1}{2} \right)$
 (c) 0
 (d) $nRT \ln 2$

17. If the process on one mole of monatomic ideal gas is as shown in the TV -diagram with $P_0 V_0 = \frac{1}{3} RT_0$, the correct match is [JEE Advance 2019]

List-I		List-II	
A.	Work done by the system in process energy $1 \rightarrow 2 \rightarrow 3$	I.	$\frac{1}{3} RT_0 \ln 2$
B.	Change in internal in process $1 \rightarrow 2 \rightarrow 3$	II.	$\frac{1}{3} RT_0$
C.	Heat absorbed by the system in process $1 \rightarrow 2 \rightarrow 3$	III.	RT_0
D.	Heat absorbed by the system in process $1 \rightarrow 2$	IV.	$\frac{4}{3} RT_0$
		V.	$\frac{1}{3} RT_0 (3 + \ln 2)$
		VI.	$\frac{5}{6} RT_0$



- (a) $A \rightarrow IV$; $B \rightarrow V$; $C \rightarrow II$; $D \rightarrow VI$
 (b) $A \rightarrow I$; $B \rightarrow III$; $C \rightarrow V$; $D \rightarrow IV$
 (c) $A \rightarrow I$; $B \rightarrow V$; $C \rightarrow II$; $D \rightarrow V$
 (d) $A \rightarrow I$; $B \rightarrow III$; $C \rightarrow V$; $D \rightarrow I$
18. n -moles of an ideal gas with constant volume heat capacity C_v undergo an isobaric expansion by certain volume. The ratio of the work done in the process, to the heat supplied is [10 April, 2019 (Shift-I)]

- (a) $\frac{4nR}{C_v - nR}$ (b) $\frac{nR}{C_v - nR}$
 (c) $\frac{nR}{C_v + nR}$ (d) $\frac{4nR}{C_v + nR}$

19. The pressure (P) and temperature (T) relationship of an ideal gas obeys the equation $PT^2 = \text{constant}$. The volume expansion coefficient of the gas will be: [30 Jan, 2023 (Shift-I)]

- (a) $3T$ (b) $\frac{3}{T^2}$
 (c) $\frac{3}{T^3}$ (d) $\frac{3}{T}$

20. A vessel contains 16 g of hydrogen and 128 g of oxygen at standard temperature and pressure. The volume of the vessel in cm^3 is: [29 June, 2022 (Shift-II)]

(a) 72×10^5 (b) 32×10^5
(c) 27×10^4 (d) 54×10^4

21. Three vessels of equal volume contain gases at the same temperature and pressure. The first vessel contains neon (monoatomic), the second contains chlorine (diatomic) and third contains uranium hexafluoride (polyatomic). Arrange these on the basis of their root mean square speed (V_{rms}) and choose the correct answer from the options given below: [11 Apr, 2023 (Shift-I)]

(a) $V_{rms}(\text{mono}) = V_{rms}(\text{dia}) = V_{rms}(\text{poly})$
(b) $V_{rms}(\text{mono}) > V_{rms}(\text{dia}) > V_{rms}(\text{poly})$
(c) $V_{rms}(\text{dia}) < V_{rms}(\text{poly}) < V_{rms}(\text{mono})$
(d) $V_{rms}(\text{mono}) < V_{rms}(\text{dia}) < V_{rms}(\text{poly})$

22. According to kinetic theory of gases, [27 June, 2022 (Shift-II)]

(A) The motion of the gas molecules freezes at 0°C
(B) The mean free path of gas molecules decreases if the density of molecules is increased.
(C) The mean free path of gas molecules increases if temperature is increased keeping pressure constant.
(D) Average kinetic energy per molecule per degree of freedom is $\frac{3}{2}k_B T$ (for monoatomic gases)

Choose the most appropriate answer from the options given below: [27 June, 2022 (Shift-II)]

(a) A and C only (b) B and C only
(c) A and B only (d) C and D only

23. Number of molecules in a volume of 4 cm^3 of a perfect monoatomic gas at some temperature T and at a pressure of 2 cm of mercury is close to? (Given, mean kinetic energy of a molecule (at T) is $4 \times 10^{-14} \text{ erg}$, $g = 980 \text{ cm/s}^2$, density of mercury = 13.6 g/cm^3) [5 Sep, 2020 (Shift-I)]

(a) 5.8×10^{18} (b) 4.0×10^{16}
(c) 5.8×10^{16} (d) 4.0×10^{18}

24. Let γ_1 be the ratio of molar specific heat at constant pressure and molar specific heat at constant volume of a monoatomic gas and γ_2 be the similar ratio of diatomic gas. Considering the diatomic gas molecule as a rigid rotator,

the ratio, $\frac{\gamma_1}{\gamma_2}$ is [24 Jan, 2023 (Shift-II)]

(a) $\frac{27}{35}$ (b) $\frac{35}{27}$
(c) $\frac{25}{21}$ (d) $\frac{21}{25}$

25. According to law of equipartition of energy the molar specific heat of a diatomic gas at constant volume where the molecule has one additional vibrational mode is:

[25 Jan, 2023 (Shift-II)]

(a) $\frac{9}{2}R$ (b) $\frac{5}{2}R$
(c) $\frac{3}{2}R$ (d) $\frac{7}{2}R$

26. The mean free path of molecules of a certain gas at STP is $1500d$, where d is the diameter of the gas molecules. While maintaining the standard pressure, the mean free path of the molecules at 373K is approximately:

[13 Apr, 2023 (Shift-II)]

(a) $1098d$ (b) $2049d$
(c) $750d$ (d) $1500d$

27. Calculate the value of mean free path (λ) for oxygen molecules at temperature 27°C and pressure $1.01 \times 10^5 \text{ Pa}$. Assume the molecular diameter 0.3 nm and the gas is ideal.

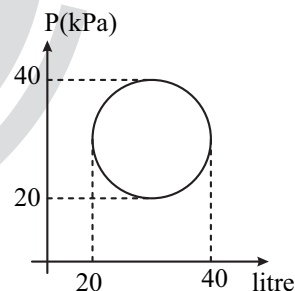
($k = 1.38 \times 10^{-23} \text{ JK}^{-1}$) [16 March, 2021 (Shift-II)]

(a) 58 nm (b) 86 nm
(c) 32 nm (d) 102 nm

Integer Type Questions

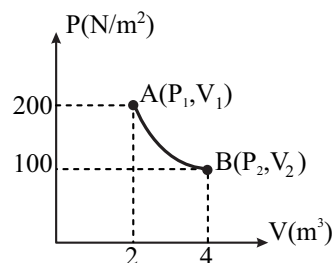
28. In the reported figure, heat energy absorbed by a system in going through a cyclic process is πJ .

[20 July, 2021 (Shift-I)]



29. One mole of an ideal gas at 27°C is taken from A to B as shown in the given PV indicator diagram. The work done by the system will be $\times 10^{-1} \text{ J}$.

[20 July, 2021 (Shift-II)]



[Given: $R = 8.3 \text{ J/mole K}$, $\ln 2 = 0.6931$]

(Round off to the nearest integer)

30. Starting at temperature 300 K, one mole of an ideal diatomic gas ($\gamma = 1.4$) is first compressed adiabatically from volume V_1 to $V_2 = V_1/16$. It is then allowed to expand isobarically to volume $2V_2$. If all the processes are the quasi-static then the final temperature of the gas (in °K) is (to the nearest integer) _____.

[9 Jan, 2020 (Shift-II)]



ANSWER KEY

1. (b) 2. (a) 3. (d) 4. (d) 5. (d) 6. (d) 7. (c) 8. (a) 9. (d) 10. (d)
 11. (b) 12. (b) 13. (c,d) 14. (a) 15. (c) 16. (b) 17. (d) 18. (c) 19. (d) 20. (c)
 21. (b) 22. (b) 23. (d) 24. (c) 25. (d) 26. (b) 27. (d) 28. [100] 29. [17258] 30. [1819]

EXPLANATIONS

1. (b) Work done = Area of triangle

$$\Rightarrow W = +\frac{1}{2}(4-2) \times (400-100)$$

$$W = 300 \text{ J}$$

2. (a) From first law of thermodynamics

$$\Delta U = \Delta Q - \Delta W$$

$$= mL_v - P\Delta V$$

$$= (1)(2257 \times 10^3) - (1 \times 10^5)(1.671 - 1 \times 10^3)$$

$$= 2257 \times 10^3 - 167 \times 10^3 = 2090 \times 10^3 \text{ J}$$

$$= 2090 \text{ kJ}$$

3. (d) If heat is added to a system (Isothermal) then temperature will remain constant. Hence statement-I is false

Statement-I: is false

Statement-II: If work done is positive then

$$PdV > 0$$

Hence volume of system must increase during positive work done by the system. So statement II is true

4. (d) As there is no exchange of heat, therefore $\Delta q = 0$

$$\Delta u = -W$$

$$\therefore W = \int PdV$$

$$\Rightarrow \Delta u = 30 \times 10^3 \times 150 \times 10^{-6}$$

$$= 4500 \times 10^{-3}$$

$$\Rightarrow \Delta u = 4.5 \text{ J}$$

5. (d) Work done = Area under curve

$$= \frac{6P_1 \times 3V_1}{2} = 9P_1V_1$$

6. (d) $dQ = dU + dW \Rightarrow dU = nC_v dT$

$$dU = 0 \text{ (for isothermal)}$$

$$\therefore U = \text{constant}$$

$$\text{Also } dQ > 0 \text{ (supplied)}$$

$$\text{Hence } dW > 0$$

7. (c) We know that 22.4 litres of He = 1 mole

$$44.4 \text{ litres of He} = \frac{44.8}{22.4} \text{ moles} = 2 \text{ mole}$$

$$\therefore n = 2 \text{ moles of He}$$

By using, $Q = nC_v \Delta T$ (for fixed capacity ($v = \text{constant}$))

$$= 2 \times \frac{R}{\gamma-1} \Delta T = \frac{2 \times R}{\frac{5}{3}-1} \times 20 = 2 \times \frac{3R}{2} \times 20$$

$$Q = 60R = 60 \times 8.3 = 498 \text{ J}$$

8. (a) $\therefore$ Here, θ is heat flows in the system when path ACB is used and a is heat flows when path ADB is used.

$$Q - W = Q' - W'$$

$$60 - 30 = Q' - 10$$

$$Q' = 40 \text{ J}$$

9. (d) $T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$

$$TV^{1/2} = T_2 (2V)^{1/2}$$

$$\therefore T_2 = \frac{T}{\sqrt{2}}$$

$$\Rightarrow W = \frac{R(T_1 - T_2)}{\gamma - 1} = \frac{R\left(T - \frac{T}{\sqrt{2}}\right)}{\frac{1}{2}} = RT(2 - \sqrt{2})$$

10. (d) For gas A : Isothermal process, $\Rightarrow PV = \text{constant}$

$$P_1 V_1 = P_2 V_2$$

$$PV = P_A (V/8)$$

$$P_A = 8P$$

For gas B :

$$\text{Adiabatic process} \Rightarrow PV^\gamma = \text{constant}$$

$$\text{For monoatomic gas } \gamma = \frac{5}{3}.$$

$$P_1 V_1^\gamma = P_2 V_2^\gamma$$

$$\frac{P_B}{P} = \left(\frac{V_1}{V_2} \right)^\gamma = \left(\frac{V}{V/8} \right)^{\frac{5}{3}} \Rightarrow P_B = 32P$$

$$\therefore \frac{P_B}{P_A} = \frac{32P}{8P} = 4$$

11. (b) $\because \Delta U = nC_v \Delta T$

For an isothermal process T remains constant

$$\Rightarrow \Delta U = 0$$

$$A \longrightarrow B$$

$$\because \Delta Q = \Delta U + \Delta W$$

For an Adiabatic process

$$\Delta Q = 0$$

$$\Rightarrow \Delta U = -\Delta W$$

Work done by a gas is positive

So ΔU is negative

$$B \longrightarrow C$$

For an Isochoric process $\Delta W = 0$

$$C \longrightarrow D$$

For Isobaric process

$$\Delta W = P \Delta V \neq 0$$

$$\Delta U = nC_v \Delta T \neq 0$$

Heat absorbed goes partly to increase internal energy and partly to do work.

12. (b) Given, $P = aV^{-3} \Rightarrow PV^3 = a$

Differentiating w.r.t to pressure

$$V^3 + 3PV^2 \frac{dV}{dP} = 0$$

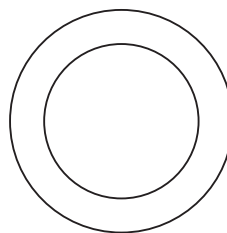
$$V = -3 \frac{PdV}{dP} = 0 \Rightarrow \frac{dP}{dV} = -3P$$

Bulk Modulus

$$B = -\frac{dP}{dV/V}$$

$$\therefore B = -\left(\frac{dPV}{dV} \right) = -(-3P) = 3P$$

13. (c,d)



$$P_{gas} = P_a + \frac{4S}{r}$$

$PV^\gamma = \text{constant}$ [adiabatic process]

$$\left(P_{a1} + \frac{4S}{r_1} \right) \left(\frac{4}{3} \pi r_1^3 \right)^{5/3} = \left(P_{a2} + \frac{4S}{r_2} \right) \left(\frac{4}{3} \pi r_2^3 \right)^{5/3}$$

$$\frac{r_1^3}{r_2^3} = \frac{\left(P_{a2} + \frac{4S}{r_2} \right)}{\left(P_{a1} + \frac{4S}{r_1} \right)}$$

$p^{1-\gamma} T^\gamma = \text{constant}$

$$\left(P_{a2} + \frac{4S}{r_2} \right)^{1-5/3} T_2^{5/3} = \left(P_{a1} + \frac{4S}{r_1} \right)^{1-5/3} T_1^{5/3}$$

$$\left(\frac{T_2}{T_1} \right)^{5/3} = \frac{\left(P_{a1} + \frac{4S}{r_1} \right)^{-2/3}}{\left(P_{a2} + \frac{4S}{r_2} \right)^{-2/3}}$$

$$\left(\frac{T_2}{T_1} \right)^{5/2} = \frac{\left(P_{a2} + \frac{4S}{r_2} \right)}{\left(P_{a1} + \frac{4S}{r_1} \right)}$$

d is correct.

14. (a) For adiabatic process

$$W = \frac{P_1 V_1 - P_2 V_2}{\gamma - 1} = \frac{100 \times 4 - 200 \times 3}{1.4 - 1} = -500 \text{ J}$$

15. (c) Given, number of mole, $n = 1$

$$W_{AB} = nRT \ln \frac{V_2}{V_1} = 1 \times RT \ln \frac{2V_1}{V_1} = RT \ln 2$$

$$W_{BC} = 0$$

$$W_{CA} = \frac{P_1 V_1 - P_2 V_2}{\gamma - 1} = \frac{\frac{P_1}{4} \times 2V_1 - P_1 \times V_1}{\gamma - 1}$$

$$\Rightarrow W_{CA} = \frac{\frac{P_1 V_1}{2} - P_1 V_1}{\gamma - 1} = \frac{-P_1 V_1}{2(\gamma - 1)} = \frac{-1RT}{2(\gamma - 1)}$$

$$\Rightarrow W_{CA} = \frac{-RT}{2(\gamma - 1)}$$

$$\therefore W_{total} = W_{AB} + W_{BC} + W_{CA} = RT \left[\ln 2 - \frac{1}{2(\gamma - 1)} \right]$$

16. (b) $W_{AB} = nRT \ln \frac{2V_1}{V_1} = nRT \ln 2$

$$W_{BC} = P_2 (V_1 - 2V_1) = -P_2 V_1 = -\frac{1}{2} nRT$$

$$[\because \text{At } B, 2P_2 V_1 = nRT]$$

As CA is isochoric process, $W_{CD} = 0$

$$W_{ABCA} = W_{AB} + W_{BC} + W_{CA} = nRT \left(\ln(2) - \frac{1}{2} \right)$$

17. (d) Process 1 $\rightarrow$ 2 is isothermal (temperature constant)
Process 2 $\rightarrow$ 3 is isochoric (volume constant)

- A. Work done in 1 $\rightarrow$ 2 $\rightarrow$ 3

$$W = W_{1 \rightarrow 2} + W_{2 \rightarrow 3}$$

$$= nRT \ln \left(\frac{V_f}{V_i} \right) + W_{2 \rightarrow 3} = \frac{RT_0}{3} \ln \left(\frac{2V_0}{V_0} \right) + 0$$

$$W = \frac{RT_0}{3} \ln 2 \Rightarrow (P)$$

- B. ΔU in 1 $\rightarrow$ 2 $\rightarrow$ 3

$$\Delta U = \frac{f}{2} nR(T_f - T_i) = \frac{3}{2} R \left(T_0 - \frac{T_0}{3} \right)$$

$$= \frac{3}{2} R \left(\frac{2T_0}{3} \right)$$

$$\Delta U = RT_0 \Rightarrow (R)$$

- C. For any system, first law of thermodynamics for 1 $\rightarrow$ 2 $\rightarrow$ 3

$$\Delta Q = \Delta U + W$$

$$\Delta Q = RT_0 + \frac{RT_0}{3} \text{ kn } 2$$

$$\Delta Q = \frac{RT_0}{3} (3 + \ln 2) \Rightarrow (T)$$

- D. For process 1 $\rightarrow$ 2 (isothermal)

$$\Delta Q = \Delta U + W$$

$$= \frac{f}{2} nR(T_f - T_i) + nRT \ln (V_f/V_i)$$

$$= 0 + R \left(\frac{T_0}{3} \right) \ln \left(\frac{2V_0}{V_0} \right)$$

$$\Delta Q = \frac{RT_0}{3} \ln 2 \Rightarrow (P)$$

18. (c) $w = nR\Delta T$

$$\Delta H = (C_v + nR)\Delta T$$

The ratio of work done to the heat supplied,

$$\frac{w}{\Delta H} = \frac{nR}{C_v + nR}$$

Hence, option (c) is correct answer.

19. (d) $PT = \text{constant}$, Using $PV = nRT$

$$P = \frac{nRT}{V}$$

$$PT^2 = \frac{nRT}{V} \times T^2 = \text{constant}$$

$$\Rightarrow T^3 = KV$$

$$\text{So, } \frac{d}{dT}(KV) = 3T^2$$

$$\Rightarrow \frac{KdV}{dT} = 3T^2$$

$$\Rightarrow dV = \frac{3T^2}{K} dT$$

$$dV = V\gamma dT$$

$$\Rightarrow \gamma V = \frac{3T^2}{K} \Rightarrow \gamma = \frac{3T^2}{KV} = \frac{3T^2}{T^3} = \frac{3}{T}$$

20. (c) No of moles of $H_2 = 8$ moles

No of moles of $O_2 = 4$ moles

Total moles = 12 moles

At STP 1 mole occupy = $22.4 \ell \times 10^3 \text{ cm}^3$

12 moles will occupy $12 \times 22.4 \times 10^3 \text{ cm}^3$

$$\approx 26.8 \times 10^4 \text{ cm}^3$$

21. (b) $V_{r.m.s} = \sqrt{\frac{3RT}{M}}$

$$\therefore V_{r.m.s} \propto \frac{1}{\sqrt{M}}$$

$$M_{\text{poly}} > M_{\text{dia}} > M_{\text{mono}}$$

$$\text{neon} = 20.17 \text{ g/mol}$$

$$Cl = 35.45 \text{ g/mol}$$

$$H.F = 146.46 \text{ g/mol}$$

So correct relation is

$$V_{r.m.s}(\text{mono}) > V_{r.m.s}(\text{dia}) > V_{r.m.s}(\text{poly})$$

$$22. (b) \lambda = \frac{kT}{\sqrt{2}\pi d^2 P}$$

$$23. (d) PV = NkT$$

$$E = \frac{3}{2}kT$$

$$N = \frac{3PV}{2E} \quad (P = hrg)$$

$$= 4 \times 10^{18}$$

$$24. (c) \text{ For a monoatomic gas } \gamma_1 = \frac{5}{3}$$

For a diatomic gas at low temperature

$$\gamma_2 = \frac{7}{5}$$

$$\therefore \frac{\gamma_1}{\gamma_2} = \frac{\frac{5}{3}}{\frac{7}{5}} = \frac{25}{21}$$

25. (d) A diatomic gas molecule has three translational degree of freedom, two rotational degree of freedom & given in the question that it has one vibrational mode so there are two additional degrees of freedom corresponding to one vibrational mode, so total degrees of freedom = 7

$$\Rightarrow C_v = \frac{7R}{2}$$

26. (b) Formula of Mean free path of a molecule of the gas

$$\lambda = \frac{RT}{\sqrt{2}\pi d^2 PN_A} \Rightarrow \lambda \propto T$$

$$\frac{1500d}{\lambda} = \frac{273}{373}$$

$$\lambda = 2049 d$$

$$27. (d) \text{ Mean free path } \lambda = \frac{kT}{1.38 \times 10^{-23} \sqrt{2} \pi d^2 P}$$

$$= \frac{1.4 \times 3.14 \times (0.3 \times 10^{-9})^2 \times 1.01 \times 10^5}{\dots}$$

$$\therefore \lambda = 102 \text{ nm}$$

28. [100] Heat absorbed in cyclic process = Work done = area of circle of radius 10 = 100π Joule

29. [17258] Given, Temperature, $T = 27^\circ\text{C} = 300$

$$V_2 = 4m^3, V_1 = 2m^3$$

Since process is isothermal, so

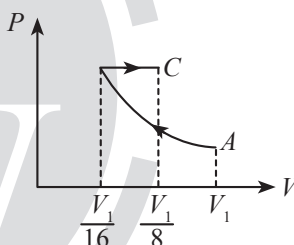
$$W = nRT \ln \left(\frac{V_2}{V_1} \right) = 1 \times 8.3 \times 300 \times \ln \left(\frac{4}{2} \right)$$

$$= 1 \times 8.3 \times 300 \times \ln(2)$$

$$\Rightarrow W = 1 \times 8.3 \times 300 \times 0.6931 \approx 1725.82 \text{ J} \approx 17258 \times 10^{-1} \text{ J}$$

30. [1819] For process AB, $PV^\gamma = \text{constant}$, $TV^{\gamma-1} = \text{constant}$

$$300(V_1)^{1.4-1} = T_B \left(\frac{V_1}{16} \right)^{0.4}$$



$$T_B = 300 \times 2^{8/5}$$

Now for BC process, pressure (P) is constant so

$$\frac{V_B}{T_B} = \frac{V_C}{T_C}$$

$$T_C = \frac{V_C T_B}{V_B} = 2 \times 300 \times 2^{8/5}$$

$$T_C = 1818.859 \approx 1819 \text{ K}$$